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Rotary position sensor with integrated potentiometer

Abstract

A rotary position sensor for measuring the position of a rotating shaft comprises an upper housing and a lower housing arranged to be attached to the upper housing. A barrel is rotationally mounted to the attached upper and lower housings and coupled to the shaft is arranged to be positionable within the attached upper and lower housings by the shaft. A potentiometer component is mounted to the barrel and to an electrical sub-assembly that measures the position of the barrel and in response generates a position signal.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure is generally directed to position sensors. More specifically, it relates to a rotary position sensor with an integrated potentiometer that generates signals representing the rotational or angular position of an actuator shaft.

BACKGROUND

(2) Rotary actuators have many applications. For example, in the automotive industry, rotary actuators may be used in valves for switching fluid paths through a heat dissipating system for cooling the various heat producing components of the vehicle. Rotary actuators can also be used to control air flow through ventilation systems used to heat and cool a vehicle's passenger cabin. In the above described applications, an electrically driven actuator motor rotates a shaft that moves a component of the controlled device connected to the shaft. For example, a rotary valve can be connected to the actuator shaft to switch fluid flow between an inlet and a plurality of outlets of the

fluid valve. The actuator motor is operated to move the rotary valve components to switch the flow of fluid between the inlet and one or more outlets using control signals sent to the actuator from control circuitry. A rotary position sensor can be used to develop and send signals back to the control circuitry as feedback confirming the position of the actuator.

SUMMARY

(3) This disclosure relates to a rotary position sensor with an integrated potentiometer that generates signals representing the rotational or angular position of an actuator shaft.

(4) In a first embodiment, a position sensor is disclosed for use with a rotating shaft. The position sensor comprises a housing and a barrel rotationally mounted in the housing coupled to the shaft and positionable in the housing by the shaft. A potentiometer component is mounted to the barrel and an electrical sub-assembly mounted to the housing is electrically connected to the potentiometer component.

(5) In a second embodiment, an assembly for measuring the position of a rotating shaft is disclosed comprising, an upper housing; a lower housing arranged to be attached to the upper housing and a barrel rotationally mounted to the attached upper and lower housings and coupled to the shaft. The barrel is positionable within the attached upper and lower housings by the shaft. A potentiometer component is mounted to the barrel and an electrical sub-assembly is mounted to the second housing and electrically connected to the potentiometer component that measures the position of the barrel and in response generates a position signal.

(6) In a third embodiment a rotary position sensor assembly for measuring the position of a rotating shaft is disclosed comprising, a lower housing and a barrel mounted to the shaft and placed in a lower housing. An upper housing is attached to the lower housing that rotatably supports the barrel between the upper and lower housings. A potentiometer component is mounted on the barrel and is rotatable with the barrel. An electrical sub-assembly is mounted to the lower housing and is electrically connected to the potentiometer component that together generate a position signal that measures the position of the barrel when the barrel is rotated by the shaft.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a perspective illustration of the rotary position sensor with an integrated potentiometer in an unassembled condition according to an embodiment of the present disclosure;

(3) FIG. 2 is a perspective illustration of the rotary position sensor with an integrated potentiometer of FIG. 1 in an assembled condition according to an embodiment of the present disclosure;

(4) FIG. 3 is a sectional illustration of the rotary position sensor with an integrated potentiometer of FIG. 2 along section lines B-B according to an embodiment of the present disclosure;

(5) FIG. 4 is a perspective illustration of the electrical sub-assembly of the rotary position sensor with an integrated potentiometer according to an embodiment of the present disclosure;

(6) FIG. 5 is a perspective illustration of the assembled rotary position sensor with an integrated potentiometer of FIG. 2 with the first and second cover portions removed to show the arrangement of the electrical sub-assembly according to an embodiment of the present disclosure; and

(7) FIG. 6 is a perspective illustration of the rotary position sensor with an integrated potentiometer in an unassembled condition according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) The figures, discussed below, and the various embodiments used to describe the principles of

the present invention in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the invention. Those skilled in the art will understand that the principles of the invention may be implemented in any type of suitably arranged device or system.

(9) FIGS. **1-6** of the present invention provides a rotary position sensor **10** for use with a mechanically or electrically driven rotating device such as for example, a rotating shaft of an actuator. The rotary position sensor **10** when attached to the shaft of an actuator (not shown) is used to sense the rotational or angular position of the shaft and to generate electrical signals using an integrated potentiometer that is indicative of its position. Control circuitry (not shown) receives the position signals as feedback signals to confirm the position of the shaft. An example of an applications using a rotary actuator includes rotary valves that use an electrical actuator to switch the flow of a fluid through the valve. The valve actuator operates to move rotary valve components to switch the flow of fluid through various passages in the valve using control signals from the control circuitry. However, it should be noted that the present invention has broad applicability and is not limited to only the application just described.

(10) FIG. **1** of the include drawings illustrates a first embodiment of an example rotary position sensor **10** of the present disclosure. The rotary position sensor **10** is comprised of an assembly having an upper housing **15**, a barrel **30**, a lower housing **40** and an electrical sub-assembly **50**. The upper housing **15**, barrel **30** and lower housing **40** may be formed as unitary structures using any thermoplastic material suitable for the application in which the sensor **10** will be used. For example, if the sensor is used in heat transfer applications using a coolant fluid, a glycol resistant thermoplastic material is used.

(11) The upper housing **15** is generally cylindrical in cross-section and includes an annular top portion **14** having a top surface **17** and a bottom surface **18**. The top portion **14** extends about the periphery of a centrally located opening **19** that extends through top surface **17** to the bottom surface **18**. The opening **19** further includes a shoulder **13** extending about opening **19**. A first arcuate outer wall **26** extend from an outer peripheral edge of the top surface **17**. A second arcuate outer wall **16** extends from the outer peripheral edge of the top surface **17** along an edge opposite from outer wall **26**. An arcuate inner wall **28** extends from the bottom surface **18** of the top portion **14**, located inward from and between outer walls **16** and **26**. A cavity **29** is defined between inner surfaces of the outer walls **16** and **26** and the inner surface of wall **28**. The cavity **29** extends axially through upper housing **15** to opening **19**. A lower portion of outer wall **26** and **16** includes a latch opening **27** that extends through each outer wall **16** and **26**. The inner wall **28** includes a wedge shaped latching member **21** extending from an outer surface of wall **28**. The top surface **17** of top portion **14** includes an aperture **88** for accepting wire conductors **57** therethrough.

(12) The lower housing **40** is generally cylindrical in cross-section and includes an annular bottom portion having a top surface **47** and a bottom surface **45**. The bottom surface **45** extends about the periphery of a centrally located opening **49** extending through top surface **47** to the bottom surface **45**. The opening **49** further includes a shoulder **41** extending about opening **49**. A first pair of arcuate outer walls **47** and **48** extend from an outer peripheral edge of the bottom portion top surface **47**. Each outer wall **47** and **48** is located along opposite edges of the bottom portion. A second pair of arcuate inner walls **46** extend from the top surface **47** of the bottom portion, located inward from and between outer walls **47** and **48**. An upper portion of outer wall **48** includes a latch opening **42** that extends through outer wall **48**. Outer wall **47** includes an opening **43** that extends through outer wall **47** and is arranged to accept portions of the electrical sub-assembly **50** therethrough. Each inner wall **46** includes a wedge shaped latching member **44** extending from an outer surface of each wall **46**.

(13) The barrel **30** includes a cylindrical wall **32** having an outer surface **34** and an inner surface **33**. The inner surface **33** encompassing a passage **81** extending axially through the barrel **30**. The barrel **30** includes a raised cylindrical potentiometer component **35** extending circumferentially

about the outer surface **34** of barrel **30**. The potentiometer component **35** located on barrel **30** so as to form an upper skirt **82** and a lower skirt **85**. An upper portion of the potentiometer component **35** forms an upper bearing surface **36** that is radially perpendicular to the upper skirt **82**. A lower portion of component **35** forms a lower bearing surface **37** that is radially perpendicular to the lower skirt **85**. Potentiometer component **35** is arranged to have electrically conductive and electrically resistive bands attached circumferentially about its outer surface. The bands include conductor bands **56** and **58** and a resistor band **57**. The resistor band **57** is electrically connected to the conductor bands **56** and **58** via end connections **55** and **59**. The bands **56**, **57** and **58** are placed on the potentiometer component in a spaced relationship to the other and are electrically interconnected by end connections **55** and **59**. Conductor band **56** is electrically connected to resistor band **57** through end connection **59**, while conductor band **58** is electrically connected to resistor band **57** via end connection **55**. With a positive electrical potential applied to conductor band **56** and a negative electrical potential applied to conductor band **58** an electrical current would flow from conductor band **56** and end connection **59** through the resistor band **57** and end connection **55** to the conductor band **58**.

(14) The conductor bands **56** and **58** may be formed from electrically conductive film and attached to component **35** using a suitable adhesive. Similarly, the resistor band **57** may be formed from an electrically resistor film and attached to component **35** also using a suitable adhesive. The end connections **55**, **59** are attached to conductive bands **56** and **57** and resistor band **57** using any appropriate electrical bridging technique, such as for example, attaching a metalized film or using solder to form an electrical bridge between resistor band **57** and the conductor bands **56**, **58**. The bands **56**, **57** and **58**, as well as the end connection **59**, **55** may also be formed directly on component **35** by depositing conductive and resistive inks directly on the outer surface of component **35** using a film deposition process.

(15) With reference to FIGS. **1**, **4** and **5** the electrical sub-assembly **50** of the present disclosure is comprised of a first cover portion **60** having an inner surface that includes a terminal mounting block **54**. The terminal mounting block **54** is arranged to receive and mount metal terminal contacts **51**, **52** and **53** on the mounting block **54**. The terminal contacts **51**, **52**, and **53** each include a bifurcated wiper portion **71** extending from a planar mounting portion **72**. The mounting portion **72** of each terminal contact **51**, **52** and **53** is attached to terminal block **54** using stakes **61** that extend from the mounting block **54** and are accepted through holes **74** on each mounting portion **72**. Each terminal contact **51**, **52** and **53** is held on the mounting block **54** by deforming each stake **61**, using any known method in the art to expand each stake **61** over its respective hole **74** to cause the expanded material of the stake **61** to capture and retain the mounting portion **72** to the mounting block **54**. Upon mounting of a terminal contact to mounting block **54** the wiper portion **71** is oriented upward and away from the mounting block **54**. A second cover portion **68** extends from an end of the first cover portion **60** and includes a wire trough formed on its interior surface by wall segments **65**. The wall segments **65** form individual compartments that house and guide each one of the wire conductor **57** to a respective terminal contact. Each conductor may be further secured to each individual compartment by applying a retaining material, such as for example and adhesive or potting compound on a bottom surface of each compartment and laying or embedding each conductor in the retaining material. Each of the wire conductors **57** connects to an individual mounting portion **72** using any suitable technique for mounting wires to electrical terminals, such as for example, a crimping sleeve. The first cover portion **60** further includes a pair of mounting holes **62** that extend through the cover to accept therein a pair of stakes **59**, that extend from the lower housing exterior wall **47**. First and second cover portions **60** and **68**, terminal mounting block **54**, stakes **61** and wall segments **65** are formed using a suitable thermoplastic material.

(16) With reference to FIGS. **1-3**, the rotary position sensor **10** is assembled by installing barrel **30** into lower housing **40** with skirt **85** through opening **49** until lower bearing surface **37** rests on shoulder **41**. With the bearing surface **37** resting the shoulder **41** the lower skirt **85** extends through

opening **49** and bottom surface **45**. The upper housing **15** is then aligned to lower housing **40**, by aligning outer walls **26** and **16** of the upper housing **15** to interior walls **46** of the lower housing **40**. As illustrated in FIGS. **1** and **3**, outer walls **26** and **16** of the upper housing **15** are arranged to slide over interior walls **46** of the lower housing **40**. Additionally, outer wall **48** of the lower housing **40** slides over inner wall **28**, thereby nesting the outer walls **26** and **16** to inner walls **46** and the outer wall **48** inner wall **28**.

(17) The upper housing **15** is pushed into the lower housing **40** until a respective latch opening **27** makes a snap-fit capture of a respective latching member **44**. Similarly latch opening **42** makes a snap-fit capture of latching member **21**. When fully installed, the barrel **30** top bearing surface **36** rotationally travels against a bottom surface of shoulder **13** with upper skirt **82** supported against shoulder **13**. The lower bearing surface **37** rotationally travels against the top surface of shoulder **41** of the lower housing **40**, as is illustrated in FIG. **3**. The upper skirt **82** centrally positions the barrel **30** in opening **19**. The upper skirt **82** extends through opening **19** with wall **32** positioned even with the top surface **17**. A bottom surface of each wall **26** and **16** rests on surface **47** of the lower housing **40** while a top surface of outer walls **47** and **48** rest against the bottom surface **18** of the upper housing **15**. The assembled upper housing **15**, barrel **30** and lower housing **40** of the rotary position sensor **10** is shown in FIG. **2** and in section drawing in FIG. **3**.

(18) As illustrated in FIG. **2**, barrel **30** is positioned to freely rotate within the installed upper and lower housings **15** and **40** along axis A in directions D. A rotational force applied to the lower skirt **85** of the barrel **30**, such as for example by installing the shaft of an actuator onto the inner surface **33** of passage **81** will rotate the barrel **30** as the actuator shaft is rotated. The lower skirt **85** may also be installed into a hollow or tubular shaft (not shown) by inserting outer surface **34** of the lower skirt **85** into the tubular shaft. A notch **87** is provided on the skirt **85** to accept a key member (not shown) extending from the actuator shaft. The notch accepting the key member therein to identify a known position on the actuator shaft.

(19) With reference to FIGS. **5**, the electrical sub-assembly **50** is shown installed into the assembled upper and lower housing **15**, **40**. FIG. **5**, is illustrated with cover portion **60** and **68** removed to illustrate the positioning of the electrical sub-assembly **50** within opening **43** of outer wall **47**. Mounting block **54** and contacts **51**, **52** and **53** extend through opening **43** toward the barrel **30** of potentiometer component **35**. Each wiper portion **71** of each terminal contact **51** and **53** makes a mechanical as well as an electrical contact with a respective band of the potentiometer component **35**. For example, the wiper portion **71** of terminal contact **52** makes an electrical contact with conductor band **56**, the wiper portion **71** of terminal contact **53** makes an electrical contact with conductor band **58** and the wiper portion **71** of terminal contact **51** makes an electrical contact with resistor band **57**. Since the wiper portions **71** of each terminal contact **51**, **52** and **53** is biased toward its respective band a mechanical spring pressure is applied by the wiper portion **71** to the band, thereby maintaining a consistent electrical connection between the terminal contact and its respective band as the barrel **30** is rotated. The second cover portion **68** with the wire conductors **57** is installed on area **88** next to inner wall **46**. The conductor wires **57** are routed through the upper housing **15** exiting aperture **88**. The first cover portion **60** and its attached second cover portion **68** are retained to the upper housing by deforming stakes **59** extending through holes **62** of the first cover portion **60**.

(20) The bands **56**, **57** and **58** of potentiometer component **35** work with the terminal contacts **52**, **51** and **53** respectively, to produce an angular or rotational potentiometer that senses the position of a rotating shaft. A direct current (DC) reference voltage applied to terminal contacts **52**, **53** via electrical wire conductors **57** are coupled to conductor bands **56** and **58** respectively via their wiper portions **71**. The DC reference voltage is electrically connected to the resistor band **57** via end connections **55** and **59** as explained above. An output voltage signal is developed by the resistor band **57** that is electrically coupled to terminal contact **51** via its wiper portion **71**. Rotational displacement of the barrel **30** causes the resistance value sensed by terminal contact **51** between the

two end connections **55**, **59** to change. The electrical resistance sensed by terminal contact **51** has a proportional relationship between the actual position of terminal contact **51** on the resistor band **57** and a resistance value. The resistance value is output as a feedback signal to a control circuit indicating the position of the barrel **30** and therefore the position of the shaft of the actuator attached to the rotary position sensor **10**.

(21) With reference to FIG. **6** an alternative embodiment is shown for attaching the upper housing **15** to the lower housing **40**. In the second embodiment a plurality of mounting stakes **120** extend from a top surface **125** of walls **146**. A plurality of mounting holes **130** axially aligned to mounting stakes **120** are located on the top portion **16** of the upper housing **15**. The mounting holes **130** extend through the top portion **16** from top surface **17** to the bottom surface **18**. The barrel **30** is installed in the lower housing **40** as was explained above and the outer walls **26** and **16** of the upper housing **15** are aligned to interior walls **146** of the lower housing **40**. Outer walls **26** and **16** of the upper housing **15** slide over interior walls **146** of the lower housing **40**. Additionally, outer wall **48** of the lower housing **40** slides over inner wall **28**, thereby nesting the outer walls **26** and **16** to inner walls **146** and the outer wall **48** to inner wall **28**.

(22) The upper housing **15** is pushed into the lower housing **40** until a respective mounting stake **120** is accepted by and extends through a respective mounting hole **130**. The mounting stakes **120** are then partially deformed using a tool or by the application of heat that attaches and retains the lower housing **40** to the upper housing **15**. The electrical sub-assembly **50** can next be installed as explained above to complete the rotary position sensor **10**.

(23) It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “communicate,” as well as derivatives thereof, encompasses both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

(24) The description in the present application should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims is intended to invoke 35 U.S.C. § 112(f) with respect to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the features of the claims themselves and is not intended to invoke 35 U.S.C. § 112(f).

(25) While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. A position sensor for a rotating shaft, comprising: a housing; a barrel rotationally mounted in the housing and coupled to the shaft and positionable in the housing by the shaft; a potentiometer component having at least a resistive band attached to an outer surface of the barrel; and an electrical sub-assembly mounted to the housing and electrically connected to the potentiometer component.
2. The position sensor of claim 1, wherein the housing comprises: an upper housing including a top portion and a pair of outer walls extending from an outer edge of the top portion and at least one inner wall located inward from and between the pair of outer walls and a cavity defined by the pair of outer walls and the at least one inner wall; and a lower housing including a bottom portion and a first outer wall extending from an outer peripheral edge of the lower housing and a pair of inner walls extending from the bottom portion located inward from the first outer wall, wherein the upper housing is arranged to be installed on the lower housing by engaging the pair of outer walls of the upper housing to the pair of inner walls of the lower housing and the first outer wall of the lower housing to the at least one inner wall of the upper housing causing the cavity to extend through the installed upper and lower housings.
3. The position sensor of claim 2, wherein: the pair of upper housing outer walls each include a latch opening and the at least one upper housing inner wall includes a latch member; and the lower housing first outer wall includes a latch opening and the pair of lower housing inner walls each include a latching member, wherein the upper housing latch member engages the lower housing latch opening, and each lower housing latch member engages a respective upper housing latch opening when the upper housing is installed on the lower housing.
4. The position sensor of claim 2, wherein the potentiometer component is attached circumferentially about the barrel outer surface forming an upper skirt and a lower skirt, the potentiometer component raised above the outer surface of the barrel creating an upper bearing surface between an upper portion of the potentiometer component and the upper skirt and a lower bearing surface between a lower portion of the potentiometer component and the lower skirt, the potentiometer component further including a pair of electrically conductive bands attached to the outer surface of the barrel.
5. The position sensor of claim 4, wherein: the upper housing top portion includes a shoulder surrounding an opening extending through the top portion and the lower housing bottom portion includes a shoulder surrounding an opening extending through the bottom portion, the potentiometer component upper bearing surface arranged to engage the upper housing shoulder with the barrel upper skirt extending to the upper housing top portion and the potentiometer component lower bearing surface arranged to engage the lower housing shoulder with the barrel lower skirt extending through and away from the bottom portion opening and coupled to the shaft, each upper and lower shoulder supporting rotational travel of the barrel within the cavity.
6. The position sensor of claim 4, wherein the electrical subassembly includes a plurality of terminal contacts and the lower housing includes a second outer wall extending from the bottom portion of the lower housing opposite the first outer wall extending to a bottom surface of the upper housing top portion, the second outer wall including an opening extending through the second outer wall to the pair of electrically conductive bands and the electrically resistive band, wherein the electrical sub-assembly is installed in the second wall opening with a respective one of the plurality of terminal contacts engaging a respective conductive band and resistive band.
7. An assembly for measuring the position of a rotating shaft comprising: an upper housing; a lower housing arranged to be attached to the upper housing; a barrel rotationally mounted to the attached upper and lower housings and coupled to the shaft, the barrel positionable within the attached upper and lower housings by the shaft; a potentiometer component having a resistive band attached to an outer surface of the barrel; and an electrical sub-assembly mounted to the lower housing and electrically connected to the potentiometer component that measures the position of the barrel and

in response generates a position signal.

8. The assembly claim 7, wherein: the upper housing includes a top portion and a pair of outer walls extending from an outer edge of the top portion and at least one inner wall located inward from and between the pair of outer walls and a cavity defined by the pair outer walls and the at least one inner wall; and the lower housing includes a bottom portion and a first outer wall extending from an outer peripheral edge of the lower housing and a pair of inner walls extending from the bottom portion located inward from the first outer wall, wherein the upper housing is installed on the lower housing by engaging the pair of outer walls of the upper housing to the pair of inner walls of the lower housing and the first outer wall of the lower housing to the at least one inner wall of the upper housing causing the cavity to extend through the installed upper and lower housings.

9. The assembly of claim 8, wherein: the pair of upper housing outer walls each include a latch opening and the at least one upper housing inner wall includes a latch member; and the lower housing first outer wall includes a latch opening and the pair of lower housing inner walls each include a latching member, wherein the upper housing latch member engages the lower housing latch opening, and each lower housing latch member engages a respective upper housing latch opening when the upper housing is installed on the lower housing retaining the upper housing to the lower housing.

10. The assembly claim 8, wherein the potentiometer component is attached circumferentially about the barrel outer surface located to form an upper skirt and a lower skirt, the potentiometer component raised above the outer surface of the barrel creating an upper bearing surface between an upper portion of the potentiometer component and the upper skirt and a lower bearing surface between a lower portion of the potentiometer component and the lower skirt, the potentiometer component including a first and a second electrically conductive band attached to the outer surface of the barrel, the resistive band electrically connected by end connections to the first and the second conductive bands.

11. The assembly of claim 10, wherein the upper housing top portion includes a shoulder surrounding an opening extending through the top portion and the lower housing bottom portion includes a shoulder surrounding an opening extending through the bottom portion, the potentiometer component upper bearing surface arranged to engage the upper housing shoulder with the barrel upper skirt extending to the upper housing top portion and the potentiometer component lower bearing surface arranged to engage the lower housing shoulder with the barrel lower skirt extending through and away from the bottom portion opening and coupled to the shaft, each upper and lower shoulder supporting the rotational travel of the barrel within the cavity.

12. The assembly of claim 10, wherein the electrical sub-assembly includes: a first cover portion having an inner surface and a mounting block formed on the first cover portion inner surface; a first, a second and a third terminal contact mounted to the mounting block, each of the first, second and third terminal contacts including a wiper portion extending from an end of each terminal contact and electrically connected to a wire conductor on an opposite end, the first and the third electrical terminal contact conductors connected to a reference voltage; a second cover portion extending from the first cover portion for housing and guiding the wire conductors to a respective terminal contact; a second outer wall extending from the bottom portion of the lower housing opposite the first outer wall extending to a bottom surface of the upper housing top portion, the second outer wall including an opening extending through the second outer wall to the first and second electrically conductive bands and the electrically resistive band, the opening receiving the mounting block to place the wiper portions of the second terminal contact in electrical contact with the resistive band and the first and the third terminal contacts in electrical contact with the first and the second conductive bands to apply the reference voltage to the first and the second conductive bands and to the resistive band through the end connections, wherein upon rotation of the barrel by the shaft the second terminal contact is moved along the resistive band to sense an output voltage

that represents the position of the barrel.

13. The assembly of claim 12, wherein the output voltage sensed by the second terminal contact is coupled by the wire conductor connected to the second terminal contact to a control circuit as a position signal representing the position of the barrel.

14. The assembly of claim 7, wherein: the upper housing includes a top portion having a plurality of holes extending through the top portion and a pair of outer walls extending from an outer edge of the top portion and at least one inner wall located inward from and between the pair of outer walls and a cavity defined by the pair outer walls and the at least one inner wall; and the lower housing include a bottom portion and a first outer wall extending from an outer peripheral edge of lower housing and a pair of inner walls extending from the bottom portion located inward from the first outer wall each of the pair of inner walls further including a plurality of stakes extending from a top surface of each inner wall, wherein the upper housing is installed on the lower housing by engaging the pair of outer walls of the upper housing to the pair of inner walls of the lower housing to insert a respective stake into a respective hole on the upper housing top portion and to install the first outer wall of the lower housing to the at least one inner wall of the upper housing causing the cavity to extend through the installed upper and lower housings.

15. The assembly claim 14, wherein each stake has a portion that extends beyond its respective hole that is deformed to retain the upper housing to the lower housing.

16. The assembly of claim 12, wherein the first cover portion includes a pair of holes extending through the first cover portion and the second outer wall includes a pair of stakes extending from the second outer wall, wherein the first cover and the second cover are attached to the lower housing by passing a respective stake through a respective one of the pair of holes and deforming a portion of the stake extending from each hole to retain the cover the lower housing.

17. A rotary position sensor assembly for measuring the position of a rotating shaft comprising: a lower housing; a barrel mounted to the shaft and placed in a lower housing; an upper housing attached to the lower housing that rotatably supports the barrel between the upper and lower housings; a potentiometer component having at least a resistive band attached on an outer surface of the barrel rotatable with the barrel; and an electrical sub-assembly mounted to the lower housing and electrically connected to the potentiometer component that together generate a position signal that measures the position of the barrel when the barrel is rotated by the shaft.

18. The rotary position sensor assembly of claim 17, wherein the potentiometer component is attached circumferentially about the barrel outer surface, the potentiometer component including a first and a second electrically conductive band attached to the outer surface of the barrel, the resistive band electrically connected by end connections to the first and the second conductive bands.

19. The rotary position sensor assembly of claim 17, wherein the electrical sub-assembly includes: a first, a second and a third terminal contact, each terminal contact including a wiper portion extending from one end and electrically connected to a wire conductor on an opposite end; the wire conductors connected to the first and the third electrical contacts connected to a reference voltage; the wiper portion of the second terminal contact electrically contacting the resistive band and the first and the third terminal contacts electrically contacting the first and the second conductive bands respectively to apply the reference voltage to the first and the second conductive bands and to the restive band through the end connections, wherein upon rotation of the barrel by the shaft, the second terminal contact is moved along the resistive band to sense an output voltage that represents the position of the barrel.

20. The assembly of claim 19, wherein the output voltage sensed by the second terminal contact is coupled by the wire conductor connected to the second terminal contact to a control circuit as a position signal representing the position of the barrel.
